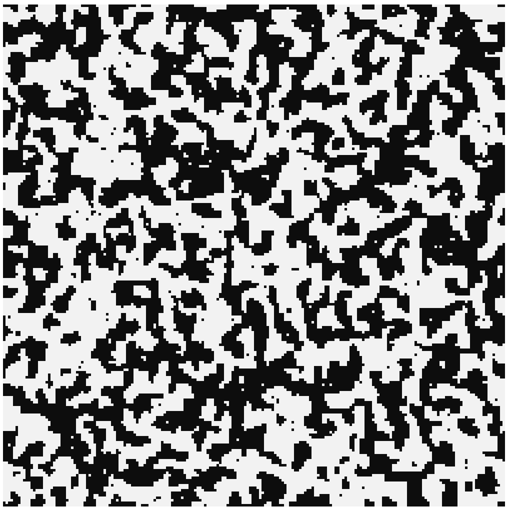
Advanced Statistical Physics

Lecture Notes

Instructor: Prof. Bhavtosh Bansal



Sagnik Seth

Dept of Physical Sciences
IISER Kolkata

Contents

Lecture 1: Introduction and Van der Waal's Equation of State

What we will discuss here in mainly the equilibrium physics of phase transitions which is both interesting and intriguing. Phase transitions are generally the result of the collective behaviour of interacting particles which produce some change in the thermodynamics of the system. For interaction between particles, we require a *pair potential*, which describes the behaviour of the interaction. One of such potentials is the celebrated *Lennard-Jones* potential (LJP) which goes as:

$$u(r) = \varepsilon_0 \left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6} \right]$$

where $r = |\mathbf{r}_i - \mathbf{r}_j|$ is the interparticle distance. This is used as a model for intermolecular interactions in many systems. Another "lesser" realistic (but simpler to handle) model can be described by the potential,

$$u(r) = \begin{cases} \infty & r < r_0 \\ -\varepsilon_0 \left(\frac{r_0}{r}\right)^6 & r > r_0 \end{cases}$$

The above potential tells that the minimum possible separation between molecules is r_0, since $u(r)$ blows up below $r = r_0$. Hence the particles act as *hard spheres* (spheres which cannot penetrate each other) of radius $\frac{r_0}{2}$. For $r > r_0$, $u(r) \sim r^{-6}$ which implies *dipolar interactions*.

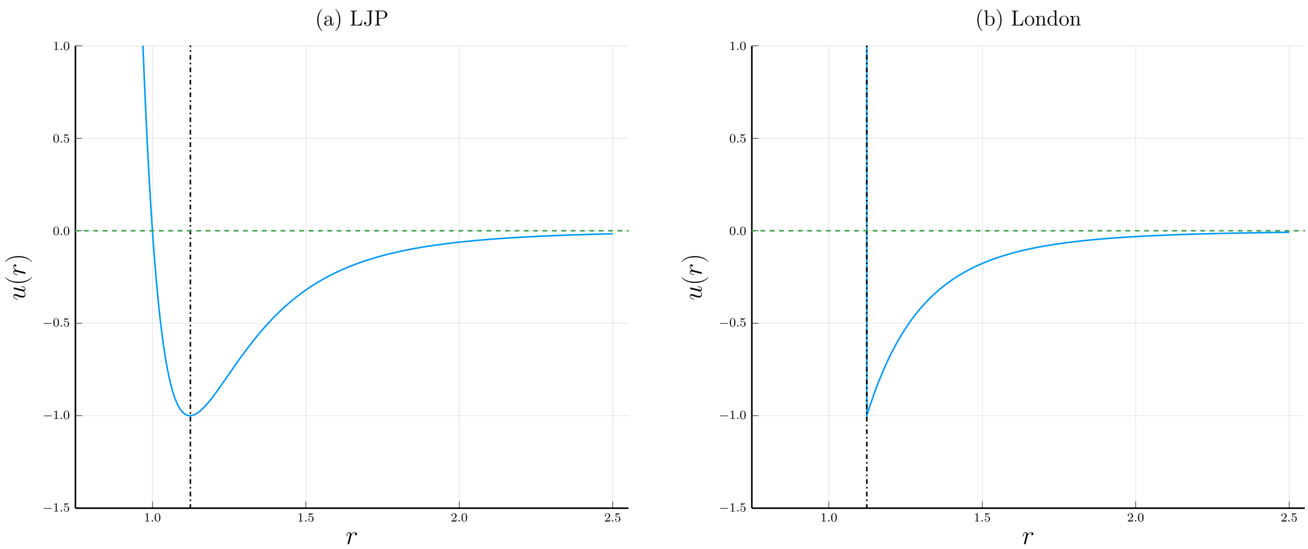


Figure 1: Interaction potentials. (a) Lennard-Jones potential. The vertical line denotes the r at which minima occurs (b) London dispersion potential.

Note that these interactions are mostly *power-law* interactions, $u(r) \sim r^{-s}$. Let d be the dimension of the space in which we are working. Then, it can be shown or motivated that if $s \leq d$, then the interaction is long-ranged which is not ideal for us. Extensivity can break down and some other issues (like negative specific heat) may occur in this case. One such example is that of gravitational systems where $u(r) \sim \frac{1}{r}$. If $s > d$ then the interaction is short-ranged. Since $u(r)$ blows up when $r = 0$, in general, it is advisable to impose a hard core repulsion (like in the London force) which ensures that everything stays okay, since interactions act only after $r > r_0$!

1.1. Van der Waal's gas

Let us see an example of this using the Van der Waal's equation which is a correction to the ideal gas equation $PV = Nk_{\mathrm{B}}T$ and successfully explains the condensation process of gases.

We will work in canonical ensemble here. The *canonical partition function* is written as,

$$\mathcal{Z}(N, V, T) = \frac{1}{N!} \int \prod_i \left(\frac{\mathrm{d}^3 p_i \, \mathrm{d}^3 q_i}{h^3} \right) e^{-\beta \mathcal{H}}$$

where $\mathcal{H}$ is the Hamiltonian of the system. The $N!$ factor has been put to include the indistinguishability of the particles and h^3 factor has been put to make the phase space volume dimensionless. The Hamiltonian consists, as usual, of a kinetic and a pair potential part.

$$\mathcal{H} = \sum_i \frac{p_i^2}{2m} + \frac{1}{2}\sum_{i \neq j} u(|\mathbf{q}_i - \mathbf{q}_j|)$$

The momentum integral can be carried out nicely since this is essentially a Gaussian integral. Using independence of the kinetic terms we get,

$$\mathcal{Z}_{\mathrm{p}} = \left[\int \frac{\mathrm{d}^3 p}{h^3}\ e^{-\beta p^2/2m}\right]^N = \left[\frac{4\pi}{h^3}\int_0^\infty p^2 e^{-\beta p^2/2m}\ \mathrm{d}p\right]^N = \left[\frac{2\pi m k_{\mathrm{B}} T}{h^2}\right]^{3N/2}$$

Defining the thermal wavelength $\lambda_{\mathrm{T}} = {}^{h}/\sqrt{2\pi m k_{\mathrm{B}} T}$, we have

$$\mathcal{Z} = \frac{1}{N!\lambda_{\mathrm{T}}^{3N}}\int \prod_i \mathrm{d}^3 q_i\ e^{-\beta U(\{\mathbf{q}_i\})}$$

where $U(\{\mathbf{q}_i\})$ denotes the interaction term in the Hamiltonian. Tackling this part is extremely hard, since the interactions are between pair of particles, hence independence cannot be assumed.

1.2. Uniform Density Approximation

One of the most popular ways to deal with interactions is through *perturbation*, however, perturbations are generally taken to be *continuous* and hence, non-analyticity (an important aspect of phase transitions) cannot be effectively captured. We will use the method of *mean-field approximation*, which treats the interactions as an “effective potential” which can be found out self-consistently. To simplify our calculations, let us define the number density of the gas, in terms of the delta function,

$$n(\mathbf{q}) = \sum_{i=1}^N\ \delta^{(3)}(\mathbf{q} - \mathbf{q}_i)$$

Then the interaction term becomes,

$$\begin{aligned}
2U &= \sum_i \sum_{j \neq i} u(|\mathbf{q}_i - \mathbf{q}_j|) \\
&= \sum_i \sum_{j \neq i} \int \mathrm{d}q_1\ \delta^{(3)}(\mathbf{q}_1 - \mathbf{q}_i) \cdot u(|\mathbf{q}_1 - \mathbf{q}_j|) \\
&= \sum_i \sum_{j \neq i} \int \mathrm{d}q_1\ \delta^{(3)}(\mathbf{q}_1 - \mathbf{q}_i) \cdot \int \mathrm{d}q_2\ \delta^{(3)}(\mathbf{q}_2 - \mathbf{q}_j) \cdot u(|\mathbf{q}_1 - \mathbf{q}_2|) \\
&= \int \mathrm{d}q_1 \mathrm{d}q_2\ \sum_i \sum_{j \neq i} \delta^{(3)}(\mathbf{q}_1 - \mathbf{q}_i) \cdot\ \delta^{(3)}(\mathbf{q}_2 - \mathbf{q}_j) \cdot u(|\mathbf{q}_1 - \mathbf{q}_2|) \\
&= \int \mathrm{d}q_1 \mathrm{d}q_2\ n(\mathbf{q}_1) n(\mathbf{q}_2) u(|\mathbf{q}_1 - \mathbf{q}_2|)
\end{aligned}$$

We will use a heavily *unmotivated* approximation where we will treat the density to be uniform, independent of the spatial coordinate, that is, $n(\mathbf{r}) \equiv n$ Then,

$$U = \frac{n^2}{2}\int \mathrm{d}q_1^3 \mathrm{d}q_2^3\ u(|\mathbf{q}_1 - \mathbf{q}_2|)$$

Let us define the variables $\mathbf{R} = \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2)$ and $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ from which we have $\mathbf{r}_1 = \mathbf{R} + \frac{\mathbf{r}}{2}$ and $\mathbf{r}_2 = \mathbf{R} - \frac{\mathbf{r}}{2}$. The Jacobian of this transformation can be written as,

$$\mathbf{J} \equiv \frac{\partial(\mathbf{r}_1, \mathbf{r}_2)}{\partial \mathbf{R}, \mathbf{r}} = \begin{pmatrix} \mathbb{1} & \frac{1}{2}\mathbb{1} \\ \mathbb{1} & -\frac{1}{2}\mathbb{1} \end{pmatrix} \implies J = \det(\mathbf{J}) = -1$$

Using the new variables, the integral becomes,

$$U = \frac{n^2}{2}\int \mathrm{d}^3R\ \mathrm{d}^3r\ u(r) = \frac{n^2 V}{2}\int_{r_0}^{\infty} \mathrm{d}^3r\ u(r) = \frac{N^2}{2V}\int \mathrm{d}^3r\ u(r)$$

where we have used $n = {}^{N}\!/\!{}_{V}$ and r_0 is the hard-core cutoff. Now, assume a power-law interaction for $r > r_0$, $u(r) = -\varepsilon_0\left(\frac{r_0}{r}\right)^s$ using which we get,

$$U = -\frac{\varepsilon_0 N^2}{2V}\cdot r_0^s \int_{r_0}^{\infty} 4\pi r^2 \mathrm{d}r\ \frac{1}{r^s} = -\frac{4\pi\varepsilon_0 N^2 r_0^s}{2V(3-s)}\ r^{3-s}\Bigg|_{r_0}^{\infty}$$

As we can see, U is finite only when $3 - s < 0 \implies s > 3$ where 3 was our spatial dimension. Assuming $s > 3$, we have $r^{3-s} \to 0$ and we have

$$U = \frac{4\pi\varepsilon_0 N^2 r_0^s}{2V(3-s)} r_0^{3-s} = -\frac{4\pi\varepsilon_0 N^2 r_0^3}{2V(s-3)} = -\frac{aN^2}{V} \qquad \text{where,}\ \ a \equiv \frac{2\pi\varepsilon_0 r_0^3}{s-3}$$

We get an 'effective' interaction potential using the *uniform density approximation.* From here, the partion function now becomes,

$$\mathcal{Z} = \frac{e^{\beta a N^2/V}}{N!\lambda_{\mathrm{T}}^{3N}}\int \prod \mathrm{d}q_i$$

Now, the integral is simply not equal to V^N, since we have assumed a hard-sphere gas and each particle has some finite volume. Assume that each particle occupies a small volume ω, which cannot be occupied by the other particles. If one particle occupies volume say V, then the second particle can occupy $V-\omega$, the third occupies $V - 2\omega$ and so on. Then the integral will be just,

$$\begin{aligned}
\int \prod \mathrm{d}q_i \equiv \prod_{i=0}^{N-1}(V - i\omega) &\approx \prod (V - j\omega)(V - (N-j)\omega) \\
&= \prod V^2\left(1 - \frac{j\omega}{V}\right)\left(1 - \frac{(N-j)\omega}{V}\right) \\
&\approx V^N \prod \left[1 - \frac{(j + N - j)\omega}{V} + \mathcal{O}(({}^{\omega}\!/\!{}_{V})^2)\right] \\
&\approx V^N\left[1 - \frac{N\omega}{V}\right]^{N/2} = \left[V\left(1 - \frac{N\omega}{V}\right)^{1/2}\right]^N
\end{aligned}$$

where we have made the product over $\approx N/2$ pairs. Now, assume $N\omega \ll V$ from which we can use the approximation $(1+x)^n \approx 1 + nx$ to get,

$$\left[V\left(1 - \frac{N\omega}{V}\right)^{1/2}\right]^N \approx \left[V\left(1 - \frac{N\omega}{2V}\right)\right]^N = \left[V - \frac{N\omega}{2}\right]^N$$

We see that the available volume is reduced by a factor of $N\omega/2$. This was the final step and we can now write the partition function,

$$\boxed{\mathcal{Z}(N,V,T) = \frac{e^{\beta a N^2/V}}{N!\lambda_{\mathrm{T}}^{3N}}\left[V - \frac{N\omega}{2}\right]^N}$$

The free energy can be obtained from the partition function using $F = -k_{\mathrm{B}}T\ln\mathcal{Z}$,

$$F = -\cancel{k_{\mathrm{B}}T}\cancel{\beta}\frac{aN^2}{V} - Nk_{\mathrm{B}}T\ln\left[V - \frac{N\omega}{2}\right] + c(N,T)$$

and from here, we can find the pressure $P = -\left(\frac{\partial F}{\partial V}\right)_{N,T}$

$$P = -\frac{aN^2}{V^2} + \frac{Nk_\text{B}T}{\left(V - \frac{N\omega}{2}\right)} \implies \boxed{\left(P + \frac{aN^2}{V^2}\right)(V - Nb) = Nk_\text{B}T} \tag{1}$$

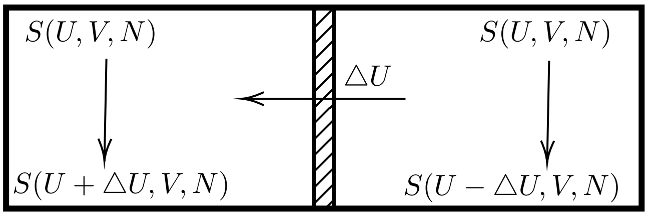
This is the Van der Waal's equation, where we have defined $b = {}^{\omega}/_{2}$. This can be written in an alternate form using the number of moles, $\mathfrak{n} = {}^{N}/_{N_\text{A}}$ where N_A is the Avogadro's number:

$$\left(P + \frac{a'\mathfrak{n}^2}{V^2}\right)(V - \mathfrak{n}b') = \mathfrak{n}RT \iff \left(P + \frac{a'}{v_m^2}\right)(v_m - b') = RT$$

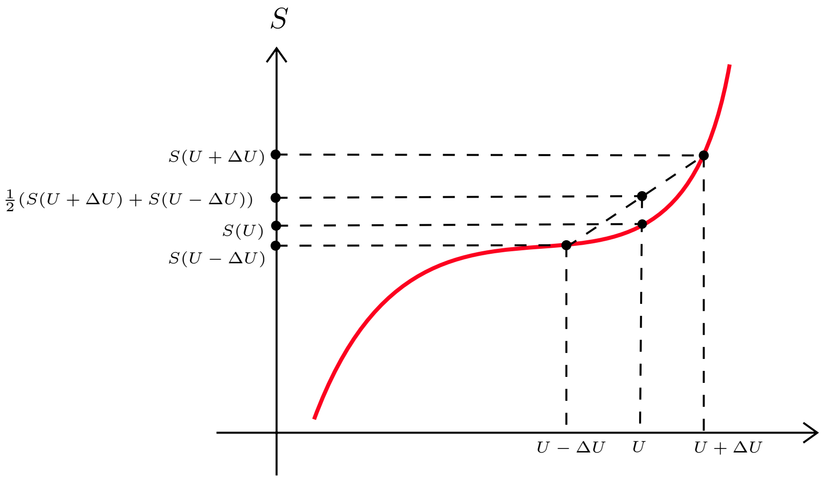
where $a' = N_\text{A}^2 a$ and $b' = bN_\text{A}$ and v_m is the molar volume.

Lecture 2: Stability of Thermodynamic Systems

We will discuss about the conditions under which the system in consideration is stable. This will in turn help us to understand *phase transitions* which are consequences of instability!



Consider two identical subsystems, with entropies $S(U, V, N)$, separated by an adiabatic wall. Suppose we remove an amount of energy ΔU from the first subsystem and transfer it to the second, the total energy changes from $2S(U, V, N)$ to $S(U + \Delta U) + S(U - \Delta U, V, N)$. Suppose that the entropy of the system looked something like in the diagram below.



Then, the resultant entropy after this transfer would be greater than the initial entropy. This would lead to inhomogenities in the system. Within one subsystem, there would be an uneven re-distribution of energy within different regions. This loss of homogenity indicates a *phase separation*!

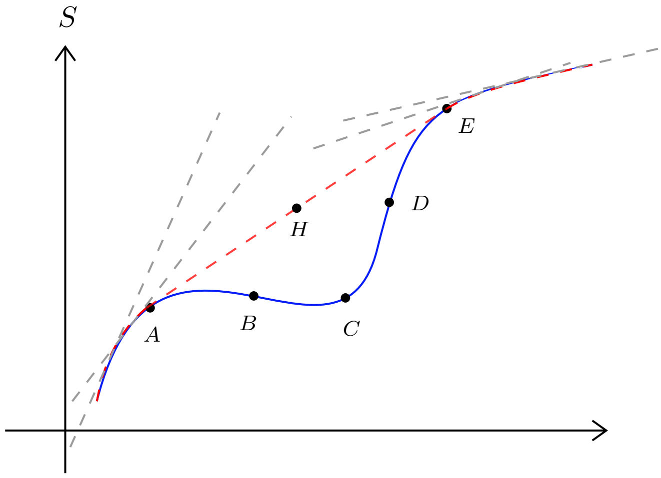
Thus the stability condition is the *concavity of entropy*, that is,

$$S(U - \Delta U, V, N) + S(U + \Delta U, V, N) \leq 2S(U) \tag{2}$$

Taylor expansion of the left hand side leads to the condition (under the limit $\Delta U \to 0$)

$$\boxed{\left(\frac{\partial^2 S}{\partial U^2}\right)_{V,N} \leq 0} \tag{3}$$

This implies that the curvature of the entropy should be downwards! It might happen that we obtain a 'locally convex' entropy curve from some calculation or from experimental data, which does not obey concavity.



This *convex intruder* cannot describe the equilibrium state and we have to obtain the *stable thermodynamic fundamental equation* from this curve. To do that, we construct all the *superior tangents*, that is, tangents which are above this curve. Now, within the family of tangents, there will be one line which will touch the curve at two places.

In the diagram, tangent at any point after A (but before E) will cross the curve. Now, the tangent at A will touch the curve at E. If it had not touched, it would have either intersected the curve or gone above E (If it has intersected the curve, then the tangent at A should not have been considered at all since we are considering only superior tangents. If it had gone over E, this means that there must be some points after A, where this touching will occur. In any case, what we want to say is that, there will be a 'common tangent' at points A and E).

Now, take the envelope of this family of tangents and this will be the fundamental thermodynamic equation that describes the equilibirum states.

Any point on the straight line denotes a phase-separated state, where part of the system is in state A and another part is in E. This happens because of the convex curve of entropy, which creates an inhomogenity.

Note that in the region AB and DE, entropy is still concave, hence these are called *metastable* or locally stable states. These states satisfy Eq.(3) but does not satisfy Eq.(2) while the region BCD satisfy neither condition.

2.1. Stability Relations for Potentials

We will now discuss what are the consequence of stability on the thermodynamic potentials, F and G. We will use the fact that for a mechanically stable system, the *specific heat* and the *thermal compressibility* must be positive for all temperatures. It can be motivated since compressibility and specific heat are related to number and energy fluctuations, σ_{N} and σ_{E} respectively, both of which are positive.

From appendix A, we see that $S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}$ from which we get:

$$\left(\frac{\partial^2 F}{\partial T^2}\right)_{V,N} = -\left(\frac{\partial S}{\partial T}\right)_{V,N} = -\frac{1}{T} C_v \leq 0$$

Similarly, we can use $S = -\left(\frac{\partial G}{\partial T}\right)_{P,N}$ from which we get:

$$\left(\frac{\partial^2 G}{\partial T^2}\right)_{P,N} = -\left(\frac{\partial S}{\partial T}\right)_{P,N} = -\frac{1}{T} C_p \leq 0$$

From this, we infer that $G(T,P,N)$ and $F(T,V,N)$ are concave functions of temperature. Apart from temperature, since F depends on V and G depends on P, let us check for pressure and volume too. We note that $V = -\left(\frac{\partial G}{\partial P}\right)_{T,N}$ and $P = -\left(\frac{\partial F}{\partial V}\right)_{T,N}$

$$\begin{aligned} \left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N} &= -\left(\frac{\partial V}{\partial P}\right)_{T,N} = -VK_{\mathrm{T}} \leq 0 \\ \left(\frac{\partial^2 F}{\partial V^2}\right)_{T,N} &= -\left(\frac{\partial P}{\partial V}\right)_{P,N} = \frac{1}{VK_{\mathrm{T}}} \geq 0 \end{aligned} \tag{4}$$

where we have appropriately used the definitions of thermal compressibility and heat capacity. Using this, we find that G is a concave function of pressure and F is a convex function of volume. From the above expressions in Eq 4, we can also write

$$\left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N} = -\left\{\left(\frac{\partial^2 F}{\partial V^2}\right)_{T,N}\right\}^{-1} \tag{5}$$

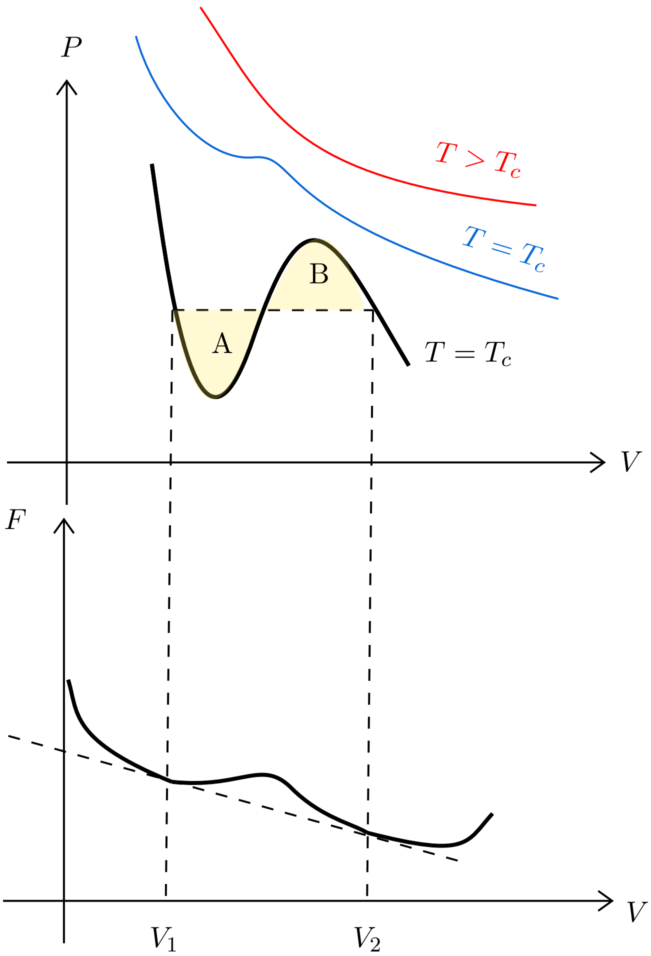
2.2. Stability Analysis of Van der waal's solutions

Recall the Van der Waal's equation from Eq (1),

$$\left(P + \frac{aN^2}{V^2}\right)(V - Nb) = Nk_{\mathrm{B}}T$$

At high temperature, the P vs. V curves at constant temperature (*isotherm*) follows the usual ideal-gas curve. However, there exists a critical temperature T_{c} (which can be found out from the equation), such that, for $T < T_{\mathrm{c}}$, there always exist a region, where the slope $\frac{\partial P}{\partial V}$ is positive. Since $K_{\mathrm{T}} = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_{N,T}$, this implies a negative compressibility, which, as motivated before, is *unphysical*.

This leads to a breakdown of the theory and Maxwell subsequently proposed an *ad hoc* remedy, similar in line to the construction of superior tangents that we noted earlier.



Consider the diagram above. Here, for a subcritical temperature, we see a region between V_1 and V_2 where the slope is positive. This region corresponds to the *concave intruder* (since free energy is a convex function of volume) in the free energy plot. Similarly as before, we replaced the concave part with a straight line denoting the commong tangent. This corresponds to joining the region between V_1 and V_2 in the P-V plot with a straight horizontal line. It can also be shown that the horizontal line is at just such a value of the pressure that the areas labelled A and B are equal and hence this construction is also called *equal-area* construction.

Lecture 3: First Order Phase Transition

Let us now see some qualitative aspect of phase transition. Each phase transition can be viewed as a result of the failure of the stability criteria. We will see some characteristic aspects of first order phase transition[1]. However, let us first note some geometric interpretation of the thermodynamic potentials.

We will consider the relation $G = F + PV$ which can be obtained from the Legendre transform of the internal energy.

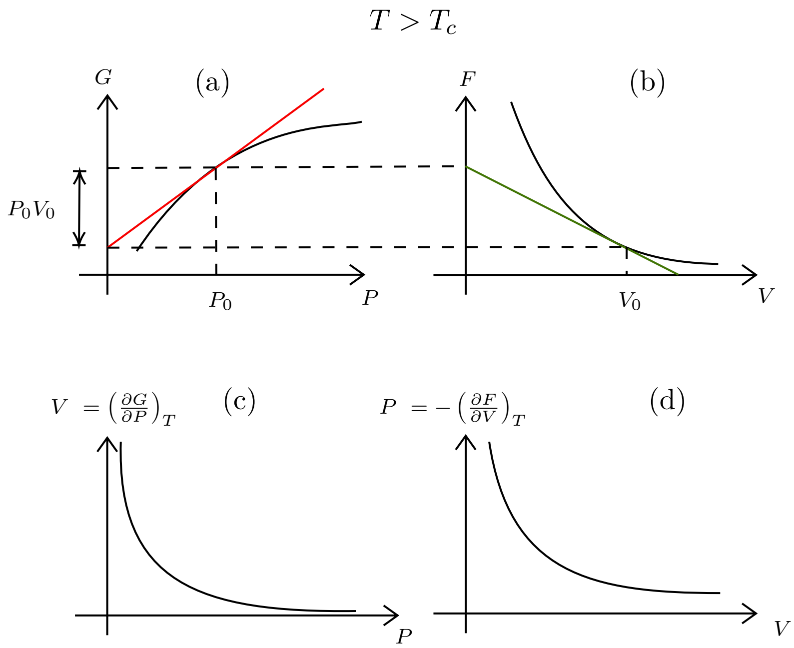


Figure 2: Relation between G and F. (a) Variation of Gibbs potential with pressure at a temperature $T > T_c$ (b) Corresponding plot of Helmholtz potential with V (c) and (d) denotes the P-V diagrams constructed from these potentials

In the above figure, observe the plot of G vs. P, which is smooth. We draw the tangent line at a $P = P_0$ and this intercepts the y-axis somewhere. The equation of the tangent is $G = PV + c$ as $V = \left(\frac{\partial G}{\partial P}\right)_{T,N}$ and it passes through the point (P_0, G_0). Thus we have,

$$G_0 = P_0V_0 + c \implies c = G_0 - P_0V_0$$

The tangent has an intercept of $G_0 - P_0V_0$ and so the remaining length as shown by the arrow in is Fig.2 (a) is $G_0 - (G_0 - P_0V_0) = P_0V_0$. From the length of the intercept of the tangent, we can obtain the Helmholtz potential for this value of P_0. We also obtain the volume from the slope of the tangent. The corresponding F vs. V graph has been plotted using these values. Also, plotted are the V vs. P graph (using the slope of the tangent of the Gibbs potential) and the P vs. V graph (using the slope of the tangent of the Helmholtz potential). This is the case for $T > T_c$. Now let us see the case when $T < T_c$ and the phase transition has occurred. We will see this with respect to the Van der Waal's equation.

[1]In modern language, these are called *abrupt phase transition*

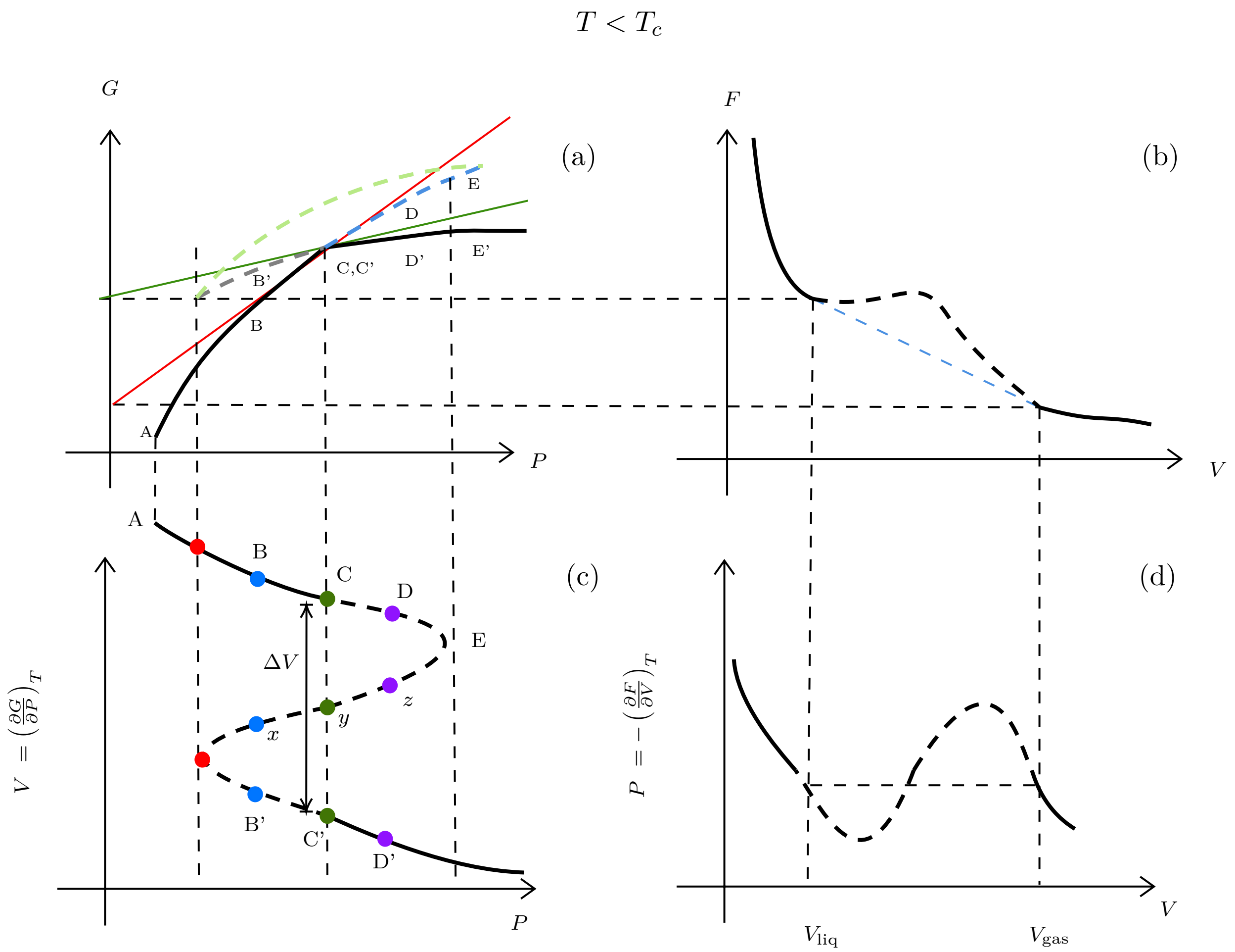


Figure 3: Plot of different quantities for $T < T_c$

In this scenario, G will have a 'kink' for some pressure $P = P_0$, since for first order phase transition, the derivative of the free energy is discontinuous. At this point, $\left(\frac{\partial^2 G}{\partial P^2}\right) \to \infty$ (even though this quantity cannot be defined exactly at $P = P_0$) and using Eq.(5), we get $\left(\frac{\partial A}{\partial V}\right) = 0$. Thus we expect that the corresponding plot of A has a *straight line* portion, as is shown in Fig.3 (b) with a blue dashed line.

Let us now see how the plot of G in (a) is actually constructed. For that, we need the isotherm in (c). Let us start at A and then, integrating we have

$$G(P) - G(P_{\text{A}}) = \int_{P_{\text{A}}}^{P} V \mathrm{d}P$$

and we do this for all values of P. This creates a closed loop kind of plot, since for a single P, we have three values of V. Note that, this disappears for higher temperature since then, Van der Waal's equation consists of only one real and two complex roots.

What happens physically? Let us suppose our system is in a temperature bath and the temperature is fixed. So, consider the isotherm in (c) and start increasing pressure from P_A. The system will follow this isotherm. At P_A, there is a unique volume and thus a unique state of the system. Now, if we increase further, for each pressure, we will get multiple values of volume, so there are different states to choose from.

Consider the blue points (all having same P but different V). Of these, the state x is unstable and hence the system has two states B and B' to choose from. The state chosen is the one with the minimum value of G which is determined from panel (a). As clearly seen, the physical state chosen is B. Increasing pressure further, we get to C where the kink is there.

On increasing further, we see that the state chosen is D' since this has lesser value of G. The blue and gray dashed lines in (a) denote the branch of G with higher value of G and thus, is not chosen. The green dashed line represent the unstable region where the states, say x, y or z lies. We see that in (c), no state from the dashed region is selected and hence after C, there is a volume jump, denoted by ΔV. The physical isotherm is thus the one created from the Maxwell equal area construction (as discussed before).

The points C and C' are determined by the condition that $G(P_c) = G(P_{c'})$ and hence, $\int\limits_{P_c}^{P_{c'}} V \mathrm{d}P = 0$, which proves that the areas divided by the line segment are indeed equal.

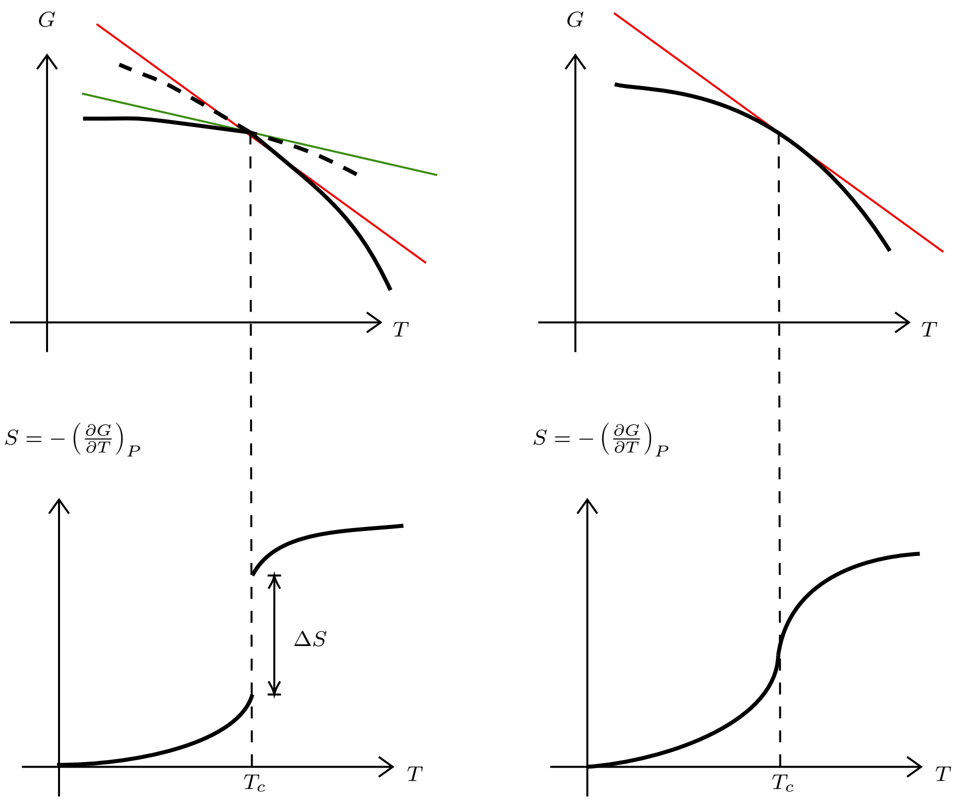


Figure 4: The left column shows the variation of G and S with temperature for first order phase transition, clearly showing an entropy discontinuity. The right column shows this for a higher order phase transition.

There is also an entropy change associated with phase transitions. For first order phase transitions, there is an abrupt entropy jump since entropy is defined as $S = -\left(\frac{\partial G}{\partial T}\right)_P$ and G has a 'kink', making it non-differentiable at the critical point. This entropy jump is associated with the latent heat $L = T_c \Delta S$. For higher order phase transitions, where G is smooth (but some higher order derivative is discontinuous), there is no latent heat since S is continuous. However, the specific heat $C \sim \frac{\mathrm{d}S}{\mathrm{d}T}$ diverges at the critical point.

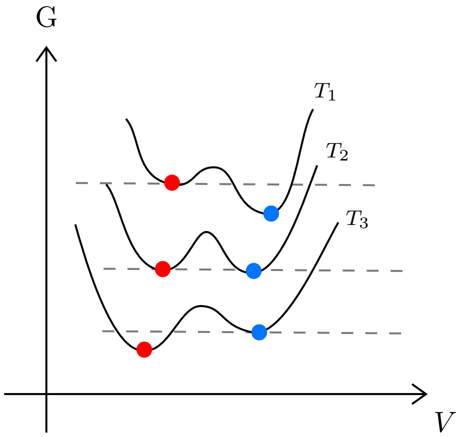


Figure 5: G vs. V plots for different temperatures.

From the previous analysis, we saw that in some interval, we had three values of volume for each pressure (out of which one was unphysical). A physical scenario to imagine is when suppose we have water above its boiling point (so it is in vapour phase). The Gibbs potential has its minima at two volumes as shown in Fig.5 for curve at T_1. The system takes the minima which is lower (the right one).

As we decrease the temperature, at some temperature, both the minima become equal, as for T_2. This is the *critical temperature*. As temperature is reduced further, the other minima becomes the more stable one. The system was in the right minima which was till now the *stable* one. It still remains in the right one, however, it now becomes *metastable*. In a very small timescale, due to some fluctuations, a *nucleus* of condensed liquid (the stable state) forms which grows rapidly and the entire system undergoes the transition.

Appendices

A. Legendre Transformation

From classical mechanics, we know that the Legendre transformation is used to convert Lagrangian $L(x, \dot{x})$ to the Hamiltonian $H(x, p)$ and vice versa. Note how we switch $\dot{x}$ in L for p in H. We will call the variable that we want to switch to as the 'conjugate' of the variable we are switching. The Legendre transformation gives us,

$$H(x, p) = L - p\dot{x}$$

We will use this for thermodynamic quantities also. From the first law, we know that,

$$U = T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}N$$

As clearly seen, the conjugate variable pairs are (S, T), $(V, -P)$, (N, μ). The internal energy $U(S, V, N)$ is a function of S, V, N from the statement of the first law. However, in any experiment, keeping these as control parameters are way too hard. Think about this, controlling entropy of a system is an arduous task but controlling its conjugate variable, T, is easy-peasy.

We get four different 'potentials' by just Legendre transformation (which is basically multiplying the conjugate variables together) of the internal energy, switching between the different conjugate variables.

- **Helmholtz Free Energy**: switching $S \leftrightarrow T$

$$F(T, V, N) = U - TS \implies \mathrm{d}F = -S\mathrm{d}T - P\mathrm{d}V + \mu\mathrm{d}N$$

- **Enthalpy**: switching $V \leftrightarrow -P$

$$H(S, P, N) = U + PV \implies \mathrm{d}H = T\mathrm{d}S + V\mathrm{d}P + \mu\mathrm{d}N$$

- **Gibb's Free Energy**: switching both $S \leftrightarrow T$ and $V \leftrightarrow -P$

$$G(T, P, N) = U - TS + PV \implies \mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}P + \mu\mathrm{d}N$$

- **Grand Potential**: switching both $S \leftrightarrow T$ and $N \leftrightarrow \mu$

$$\Phi(T, V, \mu) = U - TS - \mu N \implies \mathrm{d}\Phi = -S\mathrm{d}T - P\mathrm{d}V - N\mathrm{d}\mu$$